

# Brewster phenomenon with lossy anisotropic media

This set of notes is on the paper: Incoherent perfect absorption in lossy anisotropic materials ([https://opg.optica.org/directpdfaccess/6449ade9-eba9-401c-a5819297fdace333\\_407520/oe-27-7-9561.pdf?da=1&id=407520&seq=0&mobile=no](https://opg.optica.org/directpdfaccess/6449ade9-eba9-401c-a5819297fdace333_407520/oe-27-7-9561.pdf?da=1&id=407520&seq=0&mobile=no)) by **E.E. Narimanov**. In the below, we use the notation of this paper throughout. In **Section 1** below, we prove the **Equation 3** of the paper and also show how the results of the **Narimanov** paper reduce to the ordinary Brewster angle for an isotropic medium. In **Section 2**, we consider an anisotropic medium with loss and we derive **Equation 7** and **Equation 8** of the **Narimanov** paper, which give the required permittivity in order to have a zero reflection Brewster angle. In **Section 3**, we show that the same relation may be derived from the requirement that the dipole moments in the material be aligned with the propagation direction of the reflected field. In **Section 4**, we prove **Equation 4** of the paper (this is a rather elementary formula, which is why we relegate it to the end of these notes) and, finally, in **Section 5**, we derive **Equation 5** of the paper (well, actually, we link to another set of notes in which we had previously derived this equation).

## Section 1: Derivation of Equation 3

We consider an anisotropic medium with a diagonal permittivity tensor, where the diagonal is given by:  $(\epsilon_n, \epsilon_\tau, \epsilon_\tau)$ , where  $n, \tau$  refer to normal and tangential components (with respect to the interface). In this case, we have that the normal and tangential components of the displacement, electric, and polarization fields are given as follows:

$$D_n = \epsilon_0 \epsilon_n E_n = \epsilon_0 E_n + P_n, D_\tau = \epsilon_0 \epsilon_\tau E_\tau = \epsilon_0 E_\tau + P_\tau \rightarrow$$

$$P_n = \epsilon_0 (\epsilon_n - 1) E_n, P_\tau = \epsilon_0 (\epsilon_\tau - 1) E_\tau \rightarrow \frac{P_n}{P_\tau} = \frac{\epsilon_n - 1}{\epsilon_\tau - 1} \frac{E_n}{E_\tau} = \frac{\epsilon_n - 1}{\epsilon_\tau - 1} \frac{\epsilon_\tau}{\epsilon_n} \frac{D_n}{D_\tau},$$

which is **Equation 3** of the paper. Note that

$\nabla \cdot \mathbf{D} = 0 \rightarrow k_\tau D_\tau + k_n D_n = 0 \rightarrow D_n/D_\tau = -k_\tau/k_n$ , which is the last relation of **Equation 2** of the paper. Furthermore, note that for a lossless, isotropic medium ( $\varepsilon = \varepsilon_n = \varepsilon_\tau$ ), we have:

$$\frac{P_n}{P_\tau} = -\frac{k_\tau}{k_n} = -\frac{k_\tau}{\sqrt{\varepsilon(\omega/c)^2 - k_\tau^2}}$$

At the Brewster angle, we must have that the dipole moments are polarized along the reflection direction, which gives us the following condition:

$$\begin{aligned} \frac{k_\tau}{\sqrt{\varepsilon(\omega/c)^2 - k_\tau^2}} &= \frac{\sqrt{(\omega/c)^2 - k_\tau^2}}{k_\tau} \rightarrow \varepsilon(\omega/c)^4 - (\varepsilon + 1)(\omega/c)^2 k_\tau^2 = 0 \rightarrow k_\tau^2 = \frac{\varepsilon(\omega/c)^2}{\varepsilon + 1}, \\ \rightarrow \frac{\omega^2}{c^2} - k_\tau^2 &= \frac{\omega^2}{c^2} \frac{1}{\varepsilon + 1} \rightarrow \theta = \tan^{-1}(1/\sqrt{\varepsilon}), \end{aligned}$$

where  $\theta$  is the Brewster angle. Note that here we defined the angle to be measured from the horizontal.

## Section 2: Derivation of Equation 7 and Equation 8

In this section, we derive **Equation 7** and **Equation 8** of the paper. From **Equation 6** (which follows from setting the numerator of **Equation 4** to zero), we know that the following equality must hold for there to be no reflected wave:

$$\varepsilon_{2\tau} k_{1n} = \varepsilon_{1\tau} k_{2n} \rightarrow \varepsilon_{2\tau} k_{1n} = k_{2n} \rightarrow \varepsilon_{2\tau}'' k_{1n}' = k_{2n}'', \varepsilon_{2\tau}' k_{1n}' = k_{2n}',$$

where we have used the fact that the medium above the anisotropic substrate is vacuum (i.e.  $\varepsilon_{1\tau} = \varepsilon_{1n} \in \Re$ ). In addition, primes (double primes) indicate real (imaginary) components. Note that for a uniform medium (setting  $\varepsilon_{2\tau} = \varepsilon_{2n} = \varepsilon \in \Re$ ), we obtain:

$$\varepsilon^2 \left[ \frac{\omega^2}{c^2} - k_\tau^2 \right] = \left[ \varepsilon \frac{\omega^2}{c^2} - k_\tau^2 \right] \rightarrow \varepsilon(\varepsilon - 1)(\omega/c)^2 = (\varepsilon^2 - 1)k_\tau^2 \rightarrow \varepsilon = 1 \text{ or } k_\tau^2 = \frac{\varepsilon(\omega/c)^2}{\varepsilon + 1}$$

We now turn back to the anisotropic case. Note that the Brewster requirement of zero reflection gives us the following (using **Equation 5** and **Equation 6**):

$$\frac{k_\tau^2}{\varepsilon_{2n}} + \frac{k_{2n}^2}{\varepsilon_{2\tau}} = \frac{k_\tau^2}{\varepsilon_{2n}} + \varepsilon_{2\tau} k_{1n}^2 = k_\tau^2 \left[ \frac{1}{\varepsilon_{2n}} - \varepsilon_{2\tau} \right] + \varepsilon_{2\tau} (\omega/c)^2 = (\omega/c)^2 \rightarrow$$

$$k_\tau^2 [1 - \varepsilon_{2\tau} \varepsilon_{2n}] = [\varepsilon_{2n} - \varepsilon_{2n} \varepsilon_{2\tau}] (\omega/c)^2$$

We now take the imaginary part of this equation, which gives the following (note that  $k_\tau$  is purely real):

$$\Im \left[ \varepsilon_{2\tau} \varepsilon_{2n} \right] \left[ (\omega/c)^2 - k_\tau^2 \right] = \varepsilon_{2n}'' (\omega/c)^2 \rightarrow \frac{1}{\varepsilon_{2n}''} \Im \left[ \varepsilon_{2\tau} \varepsilon_{2n} \right] \left[ 1 - k_\tau^2 (c/\omega)^2 \right] = 1$$

$$\rightarrow k_\tau^2 = -(\omega/c)^2 \left[ \frac{\varepsilon_{2n}''}{\Im(\varepsilon_{2\tau} \varepsilon_{2n})} - 1 \right],$$

which, after replacing  $\Im(\varepsilon_{2\tau} \varepsilon_{2n}) = \varepsilon_{2\tau}' \varepsilon_{2n}'' + \varepsilon_{2\tau}'' \varepsilon_{2n}'$ , is exactly **Equation 7**. To obtain **Equation 8**, we go further by finding the corresponding real part of the equation above:

$$\Re \left[ \varepsilon_{2\tau} \varepsilon_{2n} \right] \left[ (\omega/c)^2 - k_\tau^2 \right] = \varepsilon_{2n}' (\omega/c)^2 - k_\tau^2 \rightarrow$$

$$\left[ \varepsilon_{2\tau}' \varepsilon_{2n}' - \varepsilon_{2\tau}'' \varepsilon_{2n}'' \right] \frac{\varepsilon_{2n}''}{\Im(\varepsilon_{2\tau} \varepsilon_{2n})} = (\varepsilon_{2n}' - 1) + \frac{\varepsilon_{2n}''}{\Im(\varepsilon_{2\tau} \varepsilon_{2n})},$$

where we used our expression for  $k_\tau^2$  derived above. In order to obtain an equation in the form of **Equation 8**, we make the replacement  $\Im(\varepsilon_{2\tau} \varepsilon_{2n}) = \varepsilon_{2\tau}'' \varepsilon_{2n}' + \varepsilon_{2\tau}' \varepsilon_{2n}''$  as we did before and obtain the following:

$$\left[ \varepsilon_{2\tau}' \varepsilon_{2n}' - \varepsilon_{2\tau}'' \varepsilon_{2n}'' \right] \varepsilon_{2n}'' = \varepsilon_{2n}' \left[ \varepsilon_{2n}'' \varepsilon_{2\tau}' + \varepsilon_{2n}' \varepsilon_{2\tau}'' \right] - \left[ \varepsilon_{2n}'' \varepsilon_{2\tau}' + \varepsilon_{2n}' \varepsilon_{2\tau}'' \right] + \varepsilon_{2n}'' \rightarrow$$

$$- \varepsilon_{2\tau}'' (\varepsilon_{2n}'')^2 = (\varepsilon_{2n}')^2 \varepsilon_{2\tau}'' - \left[ \varepsilon_{2n}'' \varepsilon_{2\tau}' + \varepsilon_{2n}' \varepsilon_{2\tau}'' \right] + \varepsilon_{2n}''$$

From which we get:

$$(\varepsilon_{2n}')^2 + (\varepsilon_{2n}'')^2 + \varepsilon_{2n}''/\varepsilon_{2\tau}'' - \varepsilon_{2n}' - \varepsilon_{2n}'' \varepsilon_{2\tau}'/\varepsilon_{2\tau}'' = 0,$$

which is **Equation 8**, as desired. An important point, which I don't think is made in the **Narimanov** paper is that if we have an isotropic medium with loss ( $\varepsilon' = \varepsilon'_{2n} = \varepsilon'_{2\tau}$ ,  $\varepsilon'' = \varepsilon''_{2n} = \varepsilon''_{2\tau}$ ), the above reduces to the following:

$$(\varepsilon')^2 + (\varepsilon'')^2 + 1 - 2\varepsilon' = (\varepsilon'')^2 + (\varepsilon' - 1)^2 = 0$$

Note that this is not possible for  $\varepsilon'' \neq 0$ . This is why the Brewster condition is destroyed when there is loss.

### Section 3: Consistency with phase cancelling condition

The **Narimanov** paper states that their **Equation 8** is consistent with the phase cancelling condition. I assume this means that the following quantity is purely real:

$$\frac{\varepsilon_{2n} - 1}{\varepsilon_{2\tau} - 1} \frac{\varepsilon_{2\tau}}{\varepsilon_{2n}} \frac{k_{2\tau}}{k_{2n}} = \frac{k_{1n}}{k_{1\tau}},$$

which is a generalization of the equation we had in **Section 1**. Moving various terms around, we obtain the following equivalent condition:

$$k_{\tau}^2 \varepsilon_{2\tau} [\varepsilon_{2n} - 1] = k_{1n} k_{2n} \varepsilon_{2n} [\varepsilon_{2\tau} - 1]$$

We square both sides, giving the following:

$$k_{\tau}^4 \varepsilon_{2\tau}^2 [\varepsilon_{2n} - 1]^2 = \left[ \frac{\omega^2}{c^2} - k_{\tau}^2 \right] \left[ \varepsilon_{2\tau} \frac{\omega^2}{c^2} - \frac{\varepsilon_{2\tau}}{\varepsilon_{2n}} k_{\tau}^2 \right] \varepsilon_{2n}^2 [\varepsilon_{2\tau} - 1]^2$$

To solve this, we define the dimensionless quantity  $u = k_{\tau}^2 c^2 / \omega^2$  and rewrite the above as follows (after multiplying both sides by  $(c/\omega)^4$ ):

$$u^2 \varepsilon_{2\tau} [\varepsilon_{2n} - 1]^2 = [1 - u] [\varepsilon_{2n}^2 - \varepsilon_{2n} u] [\varepsilon_{2\tau} - 1]^2$$

Expanding this expression gives us the following:

$$u^2 \left[ \varepsilon_{2\tau} [\varepsilon_{2n} - 1]^2 - \varepsilon_{2n} [\varepsilon_{2\tau} - 1]^2 \right] + u \left[ \varepsilon_{2n} + 1 \right] \varepsilon_{2n} [\varepsilon_{2\tau} - 1]^2 - \varepsilon_{2n}^2 [\varepsilon_{2\tau} - 1]^2 = 0 \rightarrow$$

$$u^2 \left[ \varepsilon_{2n} \varepsilon_{2\tau} - 1 \right] \left[ \varepsilon_{2n} - \varepsilon_{2\tau} \right] + u \left[ \varepsilon_{2n} + 1 \right] \varepsilon_{2n} [\varepsilon_{2\tau} - 1]^2 - \varepsilon_{2n}^2 [\varepsilon_{2\tau} - 1]^2 = 0,$$

where in the last equality we used the following for the coefficient of  $u^2$ :

$$\left[ \varepsilon_{2\tau} \left[ \varepsilon_{2n} - 1 \right]^2 - \varepsilon_{2n} \left[ \varepsilon_{2\tau} - 1 \right]^2 \right] = \varepsilon_{2\tau} \varepsilon_{2n}^2 - \varepsilon_{2n} \varepsilon_{2\tau}^2 + (\varepsilon_{2\tau} - \varepsilon_{2n}) = \varepsilon_{2\tau} \varepsilon_{2n} (\varepsilon_{2n} - \varepsilon_{2\tau}) - (\varepsilon_{2n} - \varepsilon_{2\tau}) = (\varepsilon_{2n} \varepsilon_{2\tau} - 1) (\varepsilon_{2n} - \varepsilon_{2\tau})$$

Thus, our simplified quadratic equation may be factorized as follows:

$$\left[ u(\varepsilon_{2n} \varepsilon_{2\tau} - 1) + \varepsilon_{2n}(1 - \varepsilon_{2\tau}) \right] \left[ u(\varepsilon_{2n} - \varepsilon_{2\tau}) - \varepsilon_{2n}(1 - \varepsilon_{2\tau}) \right] = 0$$

(<https://hackmd.io/@aligho/HydOG6yHGe/edit?both>)

Edit both



It is obvious (by eye) that the coefficients of  $u$  and unity are precisely what we require. Furthermore, note that the coefficient of  $u$  is given by:

$$\varepsilon_{2n}(1 - \varepsilon_{2\tau}) \left[ (\varepsilon_{2n} - \varepsilon_{2\tau}) - (\varepsilon_{2n} \varepsilon_{2\tau} - 1) \right] = \varepsilon_{2n}(\varepsilon_{2\tau} - 1) \left[ \varepsilon_{2n} \varepsilon_{2\tau} - 1 + \varepsilon_{2\tau} - \varepsilon_{2n} \right] = \varepsilon_{2n}(\varepsilon_{2\tau} - 1)^2 (\varepsilon_{2n} + 1),$$

which is consistent with the coefficient of  $u$  in the original quadratic equation.

Factoring the quadratic shows that there are two possible roots, which we denote by  $u_1$  and  $u_2$ :

$$u_1 = -\varepsilon_{2n}(1 - \varepsilon_{2\tau}) / (\varepsilon_{2n} \varepsilon_{2\tau} - 1)$$

$$u_2 = \varepsilon_{2n}(1 - \varepsilon_{2\tau}) / (\varepsilon_{2n} - \varepsilon_{2\tau})$$

The first root,  $u_1$ , is clearly the solution we found by demanding that the reflection coefficient be zero (in the previous section).  $u_2$ , however, is an unphysical root, which we now show.  $u_2$  corresponds to:

$$k_{1n}(c/\omega) = \left[ \frac{\varepsilon_{2n} - \varepsilon_{2\tau} - \varepsilon_{2n} + \varepsilon_{2n} \varepsilon_{2\tau}}{\varepsilon_{2n} - \varepsilon_{2\tau}} \right]^{1/2} = \left[ \frac{\varepsilon_{2\tau}(\varepsilon_{2n} - 1)}{\varepsilon_{2n} - \varepsilon_{2\tau}} \right]^{1/2}$$

$$k_{2n}(c/\omega) = \left[ \varepsilon_{2\tau} - \frac{\varepsilon_{2\tau}}{\varepsilon_{2n}} \frac{\varepsilon_{2n} - \varepsilon_{2n} \varepsilon_{2\tau}}{\varepsilon_{2n} - \varepsilon_{2\tau}} \right]^{1/2} = \left[ \varepsilon_{2\tau} \left[ 1 - \frac{1 - \varepsilon_{2\tau}}{\varepsilon_{2n} - \varepsilon_{2\tau}} \right] \right]^{1/2} = k_{1n}$$

Putting  $k_\tau$ ,  $k_{1n}$  and  $k_{2n}$  back into the dispersion relation:

$$\frac{1 - \varepsilon_{2n}}{1 - \varepsilon_{2\tau}} \frac{\varepsilon_{2\tau}}{\varepsilon_{2n}} \frac{k_{2\tau}}{k_{2n}} = \frac{k_{1n}}{k_{1\tau}}$$

Gives the following:

$$\frac{1 - \varepsilon_{2n}}{1 - \varepsilon_{2\tau}} \frac{\varepsilon_{2\tau}}{\varepsilon_{2n}} k_{\tau}^2 = \frac{\varepsilon_{2\tau}(\varepsilon_{2n} - 1)}{\varepsilon_{2n} - \varepsilon_{2\tau}} \rightarrow \frac{\varepsilon_{2\tau}(\varepsilon_{2n} - 1)}{\varepsilon_{2n} - \varepsilon_{2\tau}} = - \frac{\varepsilon_{2\tau}(\varepsilon_{2n} - 1)}{\varepsilon_{2n} - \varepsilon_{2\tau}}$$

Thus, this root is clearly unphysical.

## Section 4: Derivation of Equation 4

We consider an incident field with wavenumber  $(k_{\tau}, -k_{n1})$ , a reflected field with wavenumber  $(k_{\tau}, k_{n1})$  and a transmitted field with wavenumber  $(k_{\tau}, -k_{n2})$ . We are considering TM (p-polarized) light, which means that the magnetic field is parallel to the interface and the polarization vectors of the electric field are given by

$(k_{1n}/\varepsilon_{1\tau}, k_{\tau}/\varepsilon_{1n})$  for the incident field,  
 $r_p(-k_{1n}/\varepsilon_{1\tau}, k_{\tau}/\varepsilon_{1n})$  for the reflected field,  
 $t_p(k_{2n}/\varepsilon_{2\tau}, k_{\tau}/\varepsilon_{2n})$  for the transmitted field,

where we are ignoring any normalization constants (since we are only concerned with finding the reflected field) and the form of the polarization vectors is set by the requirement  $\nabla \cdot \mathbf{D} = 0$  everywhere. The boundary conditions we use are continuity of the tangential component of the electric field and continuity of the normal component of the displacement field:

$$(k_{1n}/\varepsilon_{1\tau})(1 - r_p) = t_p(k_{2n}/\varepsilon_{2\tau})$$

$$k_{\tau}(1 + r_p) = k_{\tau}t_p$$

Combining these two equations gives us:

$$(k_{1n}/\varepsilon_{1\tau})(1 - r_p) = (1 + r_p)(k_{2n}/\varepsilon_{2\tau}) \rightarrow$$

$$r_p = \frac{k_{2n}/\varepsilon_{2\tau} - k_{1n}/\varepsilon_{1\tau}}{-k_{1n}/\varepsilon_{1\tau} - k_{2n}/\varepsilon_{2\tau}} = \frac{k_{1n}\varepsilon_{2\tau} - k_{2n}\varepsilon_{1\tau}}{k_{1n}\varepsilon_{2\tau} + k_{2n}\varepsilon_{1\tau}},$$

which is **Equation 4**.

## Section 5: Derivation of Equation 5

We do not explicitly derive **Equation 5** of the **Narimanov** paper, since we have essentially done this before anyway. See the previous set of notes: [Sipe paper on hyperbolic metamaterials](#)

(<https://hackmd.io/@aligho/Hkdnjs3cWe>) (a pdf of this set of notes may be found in the **Miscellaneous Notes** Github repo—in the metamaterials directory).